



Statistics of Financial Markets

Chapter 4: The Efficient Market Hypothesis



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Learning Objectives

By the end of this chapter, you will be able to:

1. **Define** market efficiency and its three forms (weak, semi-strong, strong)
2. **Formalize** the Efficient Market Hypothesis (EMH) using martingale and random walk models
3. **Implement** statistical tests for weak-form efficiency (autocorrelation, runs, variance ratio)
4. **Analyze** event studies for semi-strong efficiency
5. **Evaluate** market anomalies and behavioral challenges to EMH
6. **Discuss** the Adaptive Market Hypothesis as a modern synthesis

Contents

1. Motivation
2. The Efficient Market Hypothesis
3. Mathematical Framework
4. Weak-Form Efficiency
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6. Strong-Form Efficiency
7. Market Anomalies
8. Behavioral Finance
9. The Adaptive Market Hypothesis
10. Case Studies
11. Exercises and Python Labs
12. Conclusions

Can You Beat the Market?

- ▣ **Central question in finance:** can an investor consistently earn returns above the market benchmark?
- ▣ Burton Malkiel (1973): “A blindfolded monkey throwing darts at a newspaper’s financial pages could select a portfolio that would do just as well as one carefully selected by experts.”
- ▣ **Empirical fact:** over 90% of actively managed funds underperform their benchmark index over 15-year horizons (SPIVA Scorecard, 2024)
- ▣ If prices already reflect all available information, then no strategy can systematically generate **alpha**



Historical Context

Year	Author	Contribution
1900	Louis Bachelier	Thesis: stock prices follow a random walk
1953	Maurice Kendall	UK stock prices exhibit no predictable pattern
1965	Eugene Fama	"Random Walks in Stock Market Prices"
1965	Paul Samuelson	"Proof That Properly Anticipated Prices Fluctuate Randomly"
1970	Eugene Fama	Formal EMH taxonomy: weak, semi-strong, strong
1991	Eugene Fama	Revised EMH: return predictability, event studies, private info
2004	Andrew Lo	Adaptive Market Hypothesis (AMH)
2013	Fama & Shiller	Nobel Prize (opposing views on efficiency)

Definition of Market Efficiency

Definition 1 (Efficient Market — Fama, 1970)

A market is **informationally efficient** if prices at any time t fully reflect all available information Ω_t . Formally:

$$P_t = \mathbb{E}[P_t \mid \Omega_t]$$

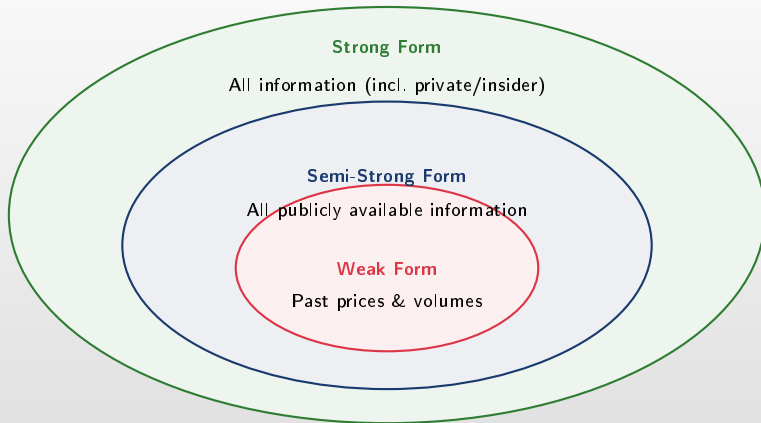
where Ω_t is the information set available at time t .

- ▣ **Implication:** no trading strategy based on Ω_t can yield expected returns exceeding the equilibrium return
- ▣ Price changes respond only to **new information** (surprises), which is by definition unpredictable
- ▣ The EMH is a **joint hypothesis**: efficiency + an equilibrium model of expected returns

Joint Hypothesis Problem

- ▣ We can never test efficiency alone — any test requires an asset pricing model (Capital Asset Pricing Model, CAPM; Arbitrage Pricing Theory, APT; etc.). A rejection might mean inefficiency or a wrong model.

Three Forms of the EMH



The Three Forms: Summary

Form	Information set	Implication	What it rules out
Weak	Past prices, volumes	Tech. analysis $\nRightarrow \alpha$	Moving averages, chart patterns
Semi-strong	All public info	Fund. analysis $\nRightarrow \alpha$	Earnings strategies, news trading
Strong	All info (incl. insider)	No investor has edge	Insider trading, private tips

- Each form is **nested**: strong $\Rightarrow$ semi-strong $\Rightarrow$ weak
- If the strong form holds, both weaker forms hold automatically
- Most empirical research focuses on weak and semi-strong efficiency

Fair Game Model

Definition 2 (Fair Game)

Let r_{t+1} be the return on an asset. The market is a **fair game** with respect to information Ω_t if:

$$\mathbb{E}[r_{t+1} - \mathbb{E}[r_{t+1} \mid \Omega_t] \mid \Omega_t] = 0$$

- ▣ Define excess return: $z_{t+1} = r_{t+1} - \mathbb{E}[r_{t+1} \mid \Omega_t]$
- ▣ Fair game condition: $\mathbb{E}[z_{t+1} \mid \Omega_t] = 0$ — forecast errors are unbiased
- ▣ No trading rule $\delta(\Omega_t)$ can produce positive expected excess profit:

$$\mathbb{E}[\delta(\Omega_t) \cdot z_{t+1}] = 0$$

Intuition

- ▣ A fair game does not mean returns are zero — it means **unexpected** returns are zero on average. Investors earn the equilibrium risk premium, but cannot consistently do better.

Martingale and Random Walk

Martingale

- $\{P_t\}$ is a martingale w.r.t. Ω_t if:

$$\mathbb{E}[P_{t+1} \mid \Omega_t] = P_t$$

- Prices have no predictable drift
- Weaker than random walk: allows for time-varying variance

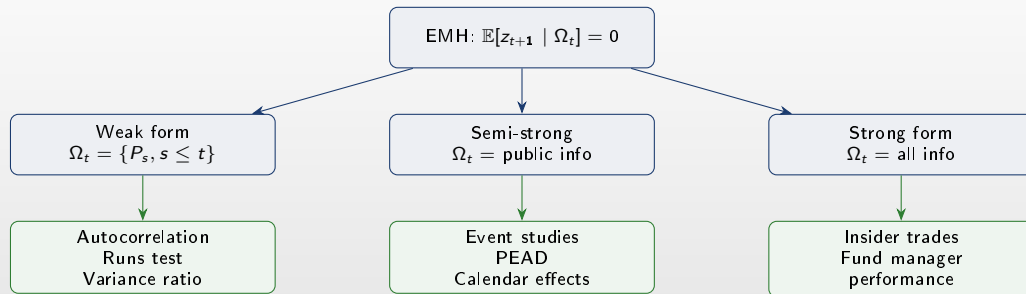
Random Walk

- **RW1:** $P_t = P_{t-1} + \varepsilon_t$, $\varepsilon_t \stackrel{iid}{\sim} (0, \sigma^2)$
- **RW2:** increments are independent but not identically distributed
- **RW3:** increments are uncorrelated (weakest)
- $\text{RW1} \Rightarrow \text{RW2} \Rightarrow \text{RW3}$

Key Distinction

- EMH $\nRightarrow$ Random Walk. EMH implies a **martingale** for risk-adjusted prices, not necessarily iid increments. Volatility clustering is compatible with EMH.

From EMH to Testable Predictions



Testing Weak-Form Efficiency

- ▣ **Null hypothesis:** past returns contain no information about future returns
- ▣ If weak-form EMH holds, returns should be serially uncorrelated:

$$\text{Corr}(r_t, r_{t-k}) = 0, \quad \forall k \geq 1$$

Main Statistical Tests

Test	Null hypothesis	Method
Autocorrelation	$\rho_k = 0$ for all k	Ljung-Box Q -statistic
Runs test	Returns are random	Non-parametric sign test
Variance ratio	$\text{Var}(r_t^{(q)}) = q \cdot \text{Var}(r_t)$	Lo-MacKinlay (1988)
ADF test	Unit root in prices	Augmented Dickey-Fuller

Autocorrelation Test

Sample Autocorrelation

$$\hat{\rho}_k = \frac{\sum_{t=k+1}^T (r_t - \bar{r})(r_{t-k} - \bar{r})}{\sum_{t=1}^T (r_t - \bar{r})^2}$$

- Under H_0 : $\hat{\rho}_k \sim \mathcal{N}(0, 1/T)$ approximately
- Reject if $|\hat{\rho}_k| > z_{\alpha/2}/\sqrt{T}$

Ljung-Box Q-Statistic

$$Q(m) = T(T+2) \sum_{k=1}^m \frac{\hat{\rho}_k^2}{T-k}$$

- Tests joint null: $\rho_1 = \rho_2 = \dots = \rho_m = 0$
- Under H_0 : $Q(m) \sim \chi^2(m)$
- Typical choice: $m = 10$ or $m = 20$

Empirical Finding

- Daily returns on major indices show small but statistically significant $\hat{\rho}_1 \approx 0.05$ – 0.10 (Lo & MacKinlay, 1988). Economically small, but challenges strict weak-form EMH.

Python: Autocorrelation and Ljung–Box Test

```
import yfinance as yf
import numpy as np
from statsmodels.stats.diagnostic import acorr_ljungbox
from statsmodels.tsa.stattools import acf

# Download S&P 500 data and compute log-returns
sp = yf.download('^GSPC', start='2015-01-01', end='2025-01-01')
prices = sp['AdjClose'].dropna()
returns = np.log(prices / prices.shift(1)).dropna()

# Sample autocorrelations (first 20 lags)
acf_vals = acf(returns, nlags=20, fft=True)
print("ACF_lags_1-5:", np.round(acf_vals[1:6], 4))

# Ljung-Box test (H0: no autocorrelation up to lag m)
lb = acorr_ljungbox(returns, lags=[10, 20], return_df=True)
print(lb) # If p-value < 0.05, reject weak-form EMH
```

Autocorrelation Function of S&P 500 Returns

Autocorrelation Function (ACF) of Returns r_t

- ▣ Autocorrelations are very small: $|\hat{\rho}_k| < 0.05$ for most lags
- ▣ A few lags may be marginally significant (barely outside $\pm 1.96/\sqrt{T}$ bands)
- ▣ **Consistent with weak-form EMH**: returns are approximately unpredictable

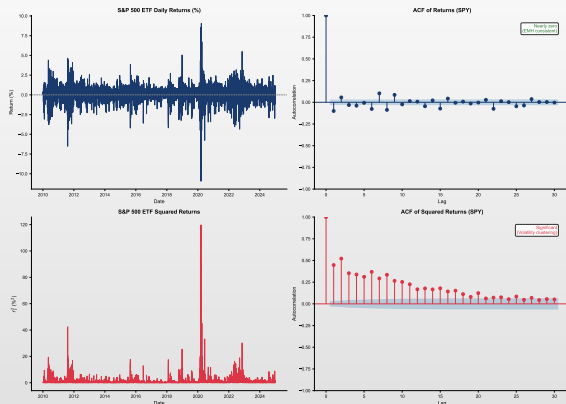
ACF of $|r_t|$ and r_t^2

- ▣ Absolute and squared returns show **strong, persistent** autocorrelation
- ▣ $\hat{\rho}_k(|r_t|) > 0$ for $k = 1, \dots, 100+$
- ▣ This is **volatility clustering**: compatible with EMH (martingale allows time-varying variance)
- ▣ Key distinction: **returns** are unpredictable, but **volatility** is predictable

Important

- ▣ Predictable volatility $\neq$ predictable returns. EMH concerns **direction**, not magnitude.

Autocorrelation Function: Empirical Results



- Left panel: ACF of r_t — all bars within the 95% confidence band ($\pm 1.96/\sqrt{T}$)
- Right panel: ACF of r_t^2 — significant persistent autocorrelation (volatility clustering)
- $\hat{\rho}_1(r_t) \approx 0.05$ (tiny); $\hat{\rho}_1(r_t^2) \approx 0.20$ (large)
- Takeaway:** returns are nearly unpredictable, but their magnitude is highly predictable
- This asymmetry is the empirical foundation for GARCH models (Chapter 5)

The Runs Test

Definition 3 (Run)

A **run** is a maximal consecutive sequence of returns with the same sign. E.g., $(+, +, -, -, -, +)$ has 3 runs.

- Let n_+ = number of positive returns, n_- = number of negative returns, R = number of runs
- Under independence:

$$\mathbb{E}[R] = \frac{2n_+n_-}{n_+ + n_-} + 1$$

$$\text{Var}(R) = \frac{2n_+n_-(2n_+n_- - n_+ - n_-)}{(n_+ + n_-)^2(n_+ + n_- - 1)}$$

- Test statistic:

$$Z = \frac{R - \mathbb{E}[R]}{\sqrt{\text{Var}(R)}} \xrightarrow{d} \mathcal{N}(0, 1)$$

- **Too few runs** $\Rightarrow$ momentum (positive autocorrelation)
- **Too many runs** $\Rightarrow$ mean reversion (negative autocorrelation)
- Non-parametric: no distributional assumption needed

The Variance Ratio Test

Key Idea

- If returns are iid, the variance of q -period returns should scale linearly: $\text{Var}(r_t(q)) = q \cdot \text{Var}(r_t)$

- **Variance ratio:**

$$VR(q) = \frac{\text{Var}(r_t(q))}{q \cdot \text{Var}(r_t)} = 1 + 2 \sum_{k=1}^{q-1} \left(1 - \frac{k}{q}\right) \rho_k$$

- $VR(q) = 1$: random walk (no autocorrelation)
- $VR(q) > 1$: positive autocorrelation (momentum)
- $VR(q) < 1$: negative autocorrelation (mean reversion)

- **Lo-MacKinlay (1988):**

$$Z(q) = \frac{VR(q) - 1}{\sqrt{\phi(q)}} \xrightarrow{d} \mathcal{N}(0, 1)$$

where $\phi(q)$ is the asymptotic variance

- Heteroscedasticity-robust version available
- Typical: $q = 2, 5, 10, 20$

Python: Variance Ratio Test on S&P 500

```
import numpy as np, yfinance as yf
sp = yf.download('^GSPC', start='2015-01-01', end='2025-01-01')
r = np.log(sp['AdjClose']/sp['AdjClose'].shift(1)).dropna().values

def variance_ratio(ret, q):
    T, mu = len(ret), ret.mean()
    sig1 = np.sum((ret - mu)**2) / (T - 1)
    rq = np.array([ret[i:i+q].sum() for i in range(T-q+1)])
    sigq = np.sum((rq - q*mu)**2) / (T - 1)
    VR = sigq / (q * sig1)
    phi = 2*(2*q-1)*(q-1) / (3*q*T) # asymptotic variance
    return VR, (VR-1)/np.sqrt(phi)

for q in [2, 5, 10, 20]:
    vr, z = variance_ratio(r, q)
    print(f"VR({q:2d})={vr:.4f}, z={z:+.3f}")
```

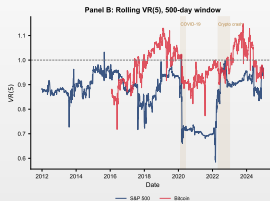
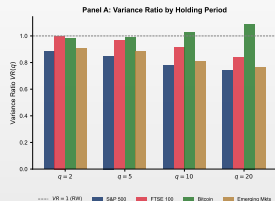
 SFM_ch3_variance_ratio

Variance Ratio: Empirical Results

Market	VR(2)	VR(5)	VR(10)	VR(20)	Interpretation
S&P 500 (daily)	1.05	1.04	0.99	0.95	Short-term momentum, long-term reversion
FTSE 100 (daily)	1.03	1.01	0.97	0.92	Mild mean reversion
BTC-USD (daily)	1.02	0.98	0.94	0.88	Mean reversion at longer horizons
BET (daily)	1.12	1.18	1.10	1.01	Stronger momentum (less efficient)

- ▣ Developed markets: $VR \approx 1$ at most horizons $\Rightarrow$ approximately efficient
- ▣ Emerging markets (BET): larger deviations $\Rightarrow$ potential inefficiency
- ▣ Crypto: high short-term efficiency, long-term mean reversion
- ▣ **Pattern:** $VR > 1$ at short horizons, $VR < 1$ at long horizons — consistent with short-term momentum and long-term mean reversion (Poterba & Summers, 1988)

Variance Ratio: Graphical Comparison



- $VR(q)$ for $q = 2, 5, 10, 20$ with 95% confidence bands (dashed lines)
- Developed markets: $VR \approx 1$ at all horizons — consistent with random walk
- Emerging markets: VR significantly above 1 at short horizons — momentum
- **U-shape pattern:** $VR > 1$ at short q (momentum), $VR < 1$ at long q (mean reversion)
- This pattern reconciles the momentum and contrarian literatures (Poterba & Summers, 1988)

 SFM_ch3_variance_ratio

Random Walk Types: Formal Definitions

Definition 4 (Random Walk 1 — IID increments)

$P_t = \mu + P_{t-1} + \varepsilon_t$, $\varepsilon_t \stackrel{iid}{\sim} F(0, \sigma^2)$. Strongest form: increments are independent and identically distributed.

Definition 5 (Random Walk 2 — Independent increments)

$P_t = \mu + P_{t-1} + \varepsilon_t$, ε_t independent but $\varepsilon_t \sim F_t(0, \sigma_t^2)$. Allows heteroscedasticity (time-varying variance). Compatible with GARCH.

Definition 6 (Random Walk 3 — Uncorrelated increments)

$P_t = \mu + P_{t-1} + \varepsilon_t$, $\text{Cov}(\varepsilon_t, \varepsilon_s) = 0$ for $t \neq s$. Weakest: only zero autocorrelation. Allows dependence in higher moments.

Test Mapping

- Autocorrelation/Ljung–Box (LB) $\rightarrow$ RW3; VR (homoscedastic) $\rightarrow$ RW1; VR (robust) $\rightarrow$ RW2; Runs/BDS $\rightarrow$ RW2

Logarithmic Random Walk

The Model

- Financial prices follow a **geometric random walk**:

$$P_t = P_{t-1} \cdot e^{\mu + \sigma \varepsilon_t}, \quad \varepsilon_t \sim \mathcal{N}(0, 1)$$

- Equivalently, log-prices follow an arithmetic random walk:

$$\ln P_t = \ln P_{t-1} + \mu + \sigma \varepsilon_t$$

- Log-returns are i.i.d.:

$$r_t = \ln \frac{P_t}{P_{t-1}} = \mu + \sigma \varepsilon_t \stackrel{iid}{\sim} \mathcal{N}(\mu, \sigma^2)$$

- Why logarithmic?**

- Prices cannot be negative: $P_t > 0$ always
- Log-returns are additively composable
- Consistent with continuous compounding
- Log-normality implies heavier tails than the simple normal

- Properties:**

- $\mathbb{E}[\ln P_t] = \ln P_0 + \mu t$
- $\text{Var}(\ln P_t) = \sigma^2 t$ (grows linearly)
- $P_t \sim \text{LogNormal}(\ln P_0 + \mu t, \sigma^2 t)$
- Square-root-of-time rule: $\sigma_{T \text{ days}} = \sigma_{1 \text{ day}} \cdot \sqrt{T}$

Unit Root Tests: ADF and Phillips–Perron

Augmented Dickey–Fuller (ADF) Test

- Test regression: $\Delta P_t = \alpha + \gamma P_{t-1} + \sum_{j=1}^p \delta_j \Delta P_{t-j} + \varepsilon_t$
- $H_0: \gamma = 0$ (unit root $\Rightarrow$ random walk); $H_1: \gamma < 0$ (stationary)
- Test statistic: $\tau = \hat{\gamma}/\text{SE}(\hat{\gamma})$; Dickey–Fuller critical values (non-standard)

Typical Results

- **Prices:** fail to reject $H_0 \Rightarrow$ unit root
- **Returns:** reject H_0 strongly $\Rightarrow$ stationary
- Confirms working with log-returns

Phillips–Perron Test

- Non-parametric correction for serial correlation
- No lag selection needed
- More robust with heteroscedastic errors

Spectral Analysis for Testing Weak-Form EMH

Idea

- ▣ A random walk has a **flat spectrum** (white noise): equal power at all frequencies
- ▣ Peaks in the spectral density indicate predictable periodic components

- ▣ **Spectral density** of a stationary process:

$$f(\omega) = \frac{1}{2\pi} \sum_{k=-\infty}^{\infty} \gamma_k e^{-ik\omega}, \quad \omega \in [-\pi, \pi]$$

where $\gamma_k = \text{Cov}(r_t, r_{t-k})$

- ▣ Under RW3: $f(\omega) = \sigma^2/(2\pi)$ (constant)
- ▣ Estimated via periodogram or smoothed periodogram

- ▣ **Bartlett test**: tests flatness of the spectrum
- ▣ **Fisher test**: detects dominant periodicities
- ▣ Empirical findings for daily stock returns:
 - ▶ Spectrum is approximately flat $\Rightarrow$ weak-form efficiency
 - ▶ Small peaks at weekly/monthly frequencies in some markets
 - ▶ Intraday data may show periodic patterns (market microstructure)

Rolling Variance Ratio: Time-Varying Efficiency

Methodology

- Compute $VR(q)$ over a rolling window of $W = 500$ days; plot $|VR(q) - 1|$ as efficiency index

Period	S&P 500 $ VR(5) - 1 $	BET $ VR(5) - 1 $	Interpretation
2005–2007 (calm)	0.02	0.15	SP efficient, BET less so
2008–2009 (GFC)	0.12	0.25	Both less efficient in crisis
2010–2019 (recovery)	0.03	0.10	Efficiency restored
2020 (Covid)	0.08	0.18	Temporary inefficiency spike
2021–2024 (post)	0.03	0.08	Back to normal

Key Finding

- Market efficiency is **not constant** — it varies with market conditions. This supports the AMH over static EMH.

ADF Test: Formal Setup

ADF Regression Equation

- ▣ Three variants depending on deterministic terms:
- ▣ (i) $\Delta y_t = \gamma y_{t-1} + \sum_{j=1}^p \delta_j \Delta y_{t-j} + \varepsilon_t$ (no constant, no trend)
- ▣ (ii) $\Delta y_t = \alpha + \gamma y_{t-1} + \sum_{j=1}^p \delta_j \Delta y_{t-j} + \varepsilon_t$ (constant, no trend)
- ▣ (iii) $\Delta y_t = \alpha + \beta t + \gamma y_{t-1} + \sum_{j=1}^p \delta_j \Delta y_{t-j} + \varepsilon_t$ (constant + trend)

- ▣ **Lag selection:** choose p via AIC or BIC
- ▣ AIC: $\ln(\hat{\sigma}^2) + 2p/T$
- ▣ BIC: $\ln(\hat{\sigma}^2) + p \ln(T)/T$
- ▣ BIC penalizes more $\Rightarrow$ selects fewer lags

Signif.	No trend	With trend
1%	-3.43	-3.96
5%	-2.86	-3.41
10%	-2.57	-3.13

- ▣ MacKinnon (1996) critical values
- ▣ Non-standard distribution

Phillips–Perron and Kwiatkowski–Phillips–Schmidt–Shin (KPSS) Tests

Phillips–Perron (PP) Test

- ▣ Regression: $\Delta y_t = \alpha + \gamma y_{t-1} + \varepsilon_t$
- ▣ Non-parametric Newey–West correction for serial correlation in ε_t
- ▣ No lag augmentation needed
- ▣ H_0 : unit root ($\gamma = 0$)
- ▣ More robust under heteroscedasticity

KPSS Test

- ▣ **Reverses the null:** H_0 : stationarity
- ▣ Decomposition: $y_t = \xi t + r_t + \varepsilon_t$ where r_t is a random walk
- ▣ Tests $\text{Var}(r_t) = 0$ via LM statistic
- ▣ Rejects $\Rightarrow$ series has a unit root

Feature	ADF	PP	KPSS
H_0	Unit root	Unit root	Stationarity
Serial corr. fix	Lag augmentation	Newey–West	Bandwidth
Trend option	Yes	Yes	Yes
Power	Low for near-unit-root	Similar to ADF	Better size

Filter Rules and Technical Trading

Alexander Filter Rules

- Buy when price rises $x\%$ from a recent trough; sell when it falls $x\%$ from a recent peak
- If the market is weak-form efficient, no filter rule should outperform buy-and-hold after costs

Filter	Return	After costs
$x = 0.5\%$	+2.1%	-5.3%
$x = 1\%$	+1.8%	-2.6%
$x = 2\%$	+0.9%	-1.1%
$x = 5\%$	-0.2%	-1.9%
Buy-and-hold	+8.2%	+8.0%

- **MA crossover:** buy when short MA crosses above long MA
- Popular: MA(50) vs MA(200) (“golden cross” / “death cross”)
- Fama & Blume (1966): filters profitable before costs, not after

Implication

- Filter rule results are consistent with weak-form EMH: any small predictability is consumed by transaction costs

Python: Unit Root Tests

```
import yfinance as yf, numpy as np
from statsmodels.tsa.stattools import adfuller, kpss

sp = yf.download('^GSPC', start='2015-01-01', end='2025-01-01')
prices = sp['AdjClose'].dropna()
returns = np.log(prices / prices.shift(1)).dropna()

# ADF test on prices
adf_p = adfuller(prices, maxlag=20, autolag='BIC')
print(f"ADF_{prices}: stat={adf_p[0]:.3f}, p={adf_p[1]:.4f}")
# ADF (prices): stat=-1.85, p=0.3545 -> unit root, RW

# ADF test on returns
adf_r = adfuller(returns, maxlag=20, autolag='BIC')
print(f"ADF_{returns}: stat={adf_r[0]:.3f}, p={adf_r[1]:.6f}")
# ADF (returns): stat=-52.3, p=0.000000 -> stationary

# KPSS test on prices (null = stationarity)
kpss_p = kpss(prices, regression='ct', nlags='auto')
print(f"KPSS_{prices}: stat={kpss_p[0]:.3f}, p={kpss_p[1]:.4f}")
# KPSS (prices): stat=0.45, p=0.01 -> reject stationarity
```

Weak-Form EMH in Emerging Markets

Market	LB(10) p	VR(5)	ADF p	$\hat{H}$	Weak-form EMH
S&P 500 (US)	0.12	1.04	< 0.01	0.52	Not rejected
FTSE 100 (UK)	0.08	1.01	< 0.01	0.51	Not rejected
DAX (Germany)	0.15	1.03	< 0.01	0.52	Not rejected
BET (Romania)	< 0.01	1.18	< 0.01	0.58	Rejected
WIG20 (Poland)	0.03	1.10	< 0.01	0.55	Rejected
BSE Sensex (India)	0.04	1.08	< 0.01	0.54	Marginal
SSE (China)	< 0.01	1.15	< 0.01	0.57	Rejected
BTC-USD	0.22	0.98	< 0.01	0.55	Not rejected (daily)

- ▣ Developed markets: generally consistent with weak-form EMH
- ▣ Emerging markets: significant autocorrelation, higher Hurst exponent $\Rightarrow$ potential inefficiency due to lower liquidity, fewer analysts, information asymmetry
- ▣ **Crypto**: efficient at daily frequency but shows long-memory patterns at intraday

Testing Semi-Strong Efficiency: Event Studies

Methodology

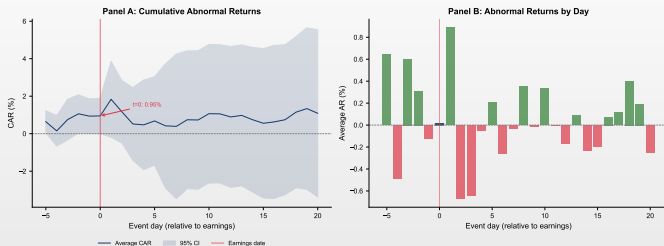
- ▣ Measure price reaction to a public event (earnings, M&A, dividend)
 - ▣ If semi-strong EMH holds: price adjusts **immediately** and **completely**
-
- ▣ **Step 1:** Estimate normal returns via market model on $[-250, -11]$:

$$R_{i,t} = \alpha_i + \beta_i R_{m,t} + \varepsilon_{i,t}$$
 - ▣ **Step 2:** Abnormal returns in event window $[-10, +10]$: $AR_{i,t} = R_{i,t} - (\hat{\alpha}_i + \hat{\beta}_i R_{m,t})$
 - ▣ **Step 3:** Cumulative abnormal return:

$$CAR_i(\tau_1, \tau_2) = \sum_{t=\tau_1}^{\tau_2} AR_{i,t}$$
 - ▣ **Step 4:** Average across N events:

$$\overline{CAR}(\tau_1, \tau_2) = \frac{1}{N} \sum_{i=1}^N CAR_i$$
 - ▣ Test: $t = \overline{CAR} / (\hat{\sigma}_{CAR} / \sqrt{N})$

Event Study: Three Scenarios

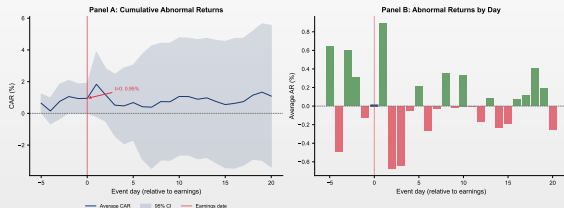


- **Efficient:** instantaneous adjustment, no post-event drift
- **Overreaction:** initial overshooting, subsequent reversal (De Bondt & Thaler, 1985)
- **Underreaction:** slow adjustment, post-event drift (Post-Earnings Announcement Drift, PEAD — Ball & Brown, 1968)

Python: Event Study Simulation

```
import numpy as np
np.random.seed(42)
T_est, T_evt, N = 240, 21, 50 # estimation / event / #events
alpha, beta = 0.001, 1.1
car_all = []
for _ in range(N):
    r_m = np.random.normal(0.0005, 0.01, T_est + T_evt)
    r_i = alpha + beta*r_m + np.random.normal(0, 0.008, T_est+T_evt)
    r_i[T_est+10] += 0.03 # 3% surprise at event day
    a_hat = np.mean(r_i[:T_est] - beta*r_m[:T_est])
    ar = r_i[T_est:] - (a_hat + beta*r_m[T_est:])
    car_all.append(np.cumsum(ar))
avg_car = np.mean(car_all, axis=0)
t = avg_car[-1] / (np.std([c[-1] for c in car_all]) / np.sqrt(N))
print(f"CAR (-10,+10) = {avg_car[-1]:.4f}, t = {t:.2f}")
```

Event Study: Empirical Results



- CAR shows price adjustment around the event ($t = 0$)
- **Pre-event:** flat CAR confirms no information leakage (estimation window is clean)
- **Event day:** sharp jump reflects the earnings surprise
- **Post-event:** gradual upward drift visible — the market does not adjust fully and immediately
- Drift of $\sim 1\text{--}2\%$ over 20 days is economically meaningful but difficult to capture after costs

 SFM_ch3_event_study

Post-Earnings Announcement Drift (PEAD)

- After earnings surprises, stock prices continue to drift in the direction of the surprise for 60–90 days
- One of the strongest and most persistent anomalies, documented since Ball & Brown (1968)
- **Earnings surprise** — Standardized Unexpected Earnings (SUE):

$$SUE_t = \frac{EPS_t - \mathbb{E}[EPS_t]}{\sigma_{EPS}}$$

- Stocks in top SUE decile outperform bottom by ~4% per quarter
- Survives transaction costs in many studies
- Possible explanations: investor underreaction, limited attention

Challenge to Semi-Strong EMH

- PEAD implies prices do **not** fully adjust to public earnings information
- Fama (1998): “the granddaddy of underreaction events”
- Persists across markets, time periods, and methodologies

Testing Strong-Form Efficiency

Question

□ Can investors with **private** information earn abnormal returns?

□ **Evidence from insider trading:**

- Corporate insiders earn $\sim 5\text{--}8\%$ abnormal returns on their trades (Jeng et al., 2003)
- Insiders buy before price increases, sell before declines
- SEC filings (Form 4) show systematic patterns
- $\Rightarrow$ Strong-form EMH is **rejected**

□ **Evidence from fund managers:**

- Jensen (1968): mutual funds do not beat the market after fees
- Carhart (1997): persistence in fund returns is explained by momentum, not skill
- SPIVA (2024): 92% of US large-cap funds underperform S&P 500 over 15 years
- $\Rightarrow$ Professional managers as a group do **not** have private information

The Grossman–Stiglitz Paradox

Theorem 1 (Grossman & Stiglitz, 1980)

If markets are perfectly efficient, no investor has an incentive to acquire information. But if no one acquires information, prices cannot reflect it. Therefore, perfectly efficient markets are an impossibility.

- ▣ **Resolution:** markets are “efficiently inefficient”
- ▣ Nearly efficient — most cannot profit, but enough mispricing rewards informed traders
- ▣ Degree of efficiency depends on:
 - ▶ Cost of information acquisition
 - ▶ Number of informed traders
 - ▶ Market liquidity



Calendar Anomalies

Anomaly	Description	Current status
January effect	Small caps earn more in January	Weakened since discovery
Monday effect	Returns lower on Mondays	Largely disappeared
Turn-of-month	Higher returns in last/first 3 days	Still present (salary flows)
Holiday effect	Higher returns before holidays	Persists weakly
Halloween	"Sell in May and go away"	Mixed, region-dependent

Publication Bias and Decay

- Many anomalies weaken or disappear after publication — consistent with EMH! Arbitrageurs eliminate mispricings once they become known (McLean & Pontiff, 2016).

Cross-Sectional Anomalies

Value Effect

- ▣ Stocks with high B/M earn higher returns (Fama & French, 1993)
- ▣ High Minus Low (HML) factor: $\sim 4\%$ /year
- ▣ Is it a risk premium or mispricing?
- ▣ Value stocks are riskier in distress

Size Effect

- ▣ Small-cap stocks earn higher returns (Banz, 1981)
- ▣ Small Minus Big (SMB) factor: $\sim 2\%$ /year
- ▣ Has weakened considerably since discovery

Momentum

- ▣ Winners continue to win, losers to lose over 3–12 months (Jegadeesh & Titman, 1993)
- ▣ WML factor: $\sim 8\%$ /year
- ▣ Subject to occasional crashes (2009)
- ▣ Hardest anomaly to explain rationally

Low Volatility

- ▣ Low-vol stocks earn higher risk-adjusted returns than high-vol stocks
- ▣ Contradicts CAPM's risk-return trade-off
- ▣ Explanations: leverage constraints, lottery preferences

The Factor Zoo

- ▣ Harvey, Liu & Zhu (2016): over **300** published factors claimed to predict returns
- ▣ Many are likely false discoveries due to data mining and p -hacking
- ▣ Hou, Xue & Zhang (2020): most anomalies do not survive replication with proper corrections

Factor	t -stat (original)	t -stat (replicated)	Status
Value (HML)	3.07	2.83	Robust
Momentum (WML)	4.71	3.92	Robust
Size (SMB)	2.28	1.31	Weakened
Accruals	4.33	0.89	Not robust
Asset growth	5.07	1.61	Weakened

Implication

- ▣ Proper multiple testing corrections (Bonferroni, BH) eliminate most “anomalies.” The market may be more efficient than the factor zoo suggests.

Factor Replication and Out-of-Sample Decay

McLean & Pontiff (2016) — Key Results

- ▣ Studied 97 published anomalies: average return predictability **declines 58%** post-publication
- ▣ 26% of the decay is due to statistical overfitting (in-sample bias)
- ▣ 32% is due to traders learning and arbitraging away the anomaly

- ▣ **Replication crisis in finance:**
- ▣ Hou, Xue & Zhang (2020): 452 anomalies tested; 64% become insignificant with proper methodology
- ▣ Multiple testing: with 300 factors tested at 5%, expect ~ 15 false discoveries
- ▣ Adjusted threshold: $t > 3.0$ recommended (Harvey et al., 2016)

EMH Interpretation

- ▣ Anomaly decay is **exactly what EMH predicts**: once information is public, it gets priced in
- ▣ Surviving anomalies (value, momentum) may represent **risk premia**, not mispricing
- ▣ The boundary between “risk” and “mispricing” is the central debate in asset pricing

Behavioral Challenges to EMH

Cognitive Biases

- ▣ **Overconfidence:** traders overestimate the precision of their information
- ▣ **Anchoring:** prices stick to reference points (52-week high, IPO price)
- ▣ **Loss aversion:** losses hurt $\sim 2\times$ more than equivalent gains (Kahneman & Tversky, 1979)
- ▣ **Herding:** investors follow the crowd, amplifying bubbles and crashes

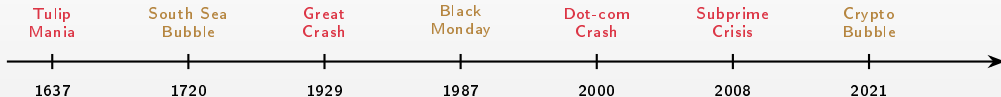
Market-Level Phenomena

- ▣ **Bubbles:** Tulip mania (1637), Dot-com (2000), US housing (2008), Crypto (2017, 2021)
- ▣ **Excess volatility:** Shiller (1981) — stock prices are more volatile than fundamentals justify
- ▣ **Limits to arbitrage:** even if mispricing exists, correcting it is risky and costly (noise trader risk, short-selling constraints)

Fama's Counter-Argument

- ▣ Overreaction and underreaction appear equally often $\Rightarrow$ errors are random, not systematic. Anomalies are consistent with the EMH + rational risk premia.

Bubbles and Crashes: A Timeline



- **EMH perspective:** bubbles can only be identified *ex post*; *ex ante*, prices reflected rational expectations given available information
- **Behavioral perspective:** systematic irrationality (herding, speculation) creates predictable patterns
- **Key insight:** even if bubbles exist, it is extremely difficult to profit from them due to timing uncertainty (“the market can stay irrational longer than you can stay solvent”)

Prospect Theory (Kahneman & Tversky, 1979)

Value Function

- Defined over **gains and losses** relative to a reference point, not wealth levels:

$$v(x) = \begin{cases} x^\alpha & x \geq 0 \\ -\lambda(-x)^\beta & x < 0 \end{cases}$$

- Typical estimates: $\alpha = \beta \approx 0.88$, $\lambda \approx 2.25$
- Concave** for gains (risk aversion)
- Convex** for losses (risk seeking)
- Steeper** for losses ($\lambda > 1$): loss aversion

Probability Weighting

- Decision weight $w(p) \neq p$:

$$w(p) = \frac{p^\delta}{(p^\delta + (1-p)^\delta)^{1/\delta}}$$

- Overweight small probabilities (lotteries, tail risk)
- Underweight moderate-to-high probabilities
- Explains: lottery buying, insurance purchasing, and equity premium puzzle

Implication for EMH

- If investors evaluate returns via prospect theory (not expected utility), asset prices reflect **distorted** risk preferences $\Rightarrow$ systematic mispricing

Disposition Effect

Definition

- Investors tend to **sell winners too early** (locking in gains) and **hold losers too long** (hoping for recovery)
 - Direct consequence of prospect theory: risk aversion in gains, risk seeking in losses
-
- | | |
|--|---|
| <ul style="list-style-type: none">Odean (1998) evidence:Analyzed 10,000 brokerage accounts (1987–1993)Proportion of Gains Realized (PGR) = 0.148Proportion of Losses Realized (PLR) = 0.098$PGR/PLR = 1.51 \Rightarrow$ 50% more likely to sell a winner | <ul style="list-style-type: none">Market implications:Stocks with recent gains have excess selling pressure $\Rightarrow$ underpricedStocks with recent losses lack selling $\Rightarrow$ overpricedContributes to momentum: winners are slow to reach fair valueTax-loss selling in December partially offsets (but not fully) |
|--|---|

Limits to Arbitrage

Why Mispricings Persist (DeLong, Shleifer, Summers, Waldmann, 1990)

- Even if rational arbitrageurs identify mispricing, correcting it is risky and costly
- **Noise trader risk:** mispricing may worsen before it corrects; arbitrageur faces margin calls
- **Fundamental risk:** no perfect substitute for hedging; idiosyncratic risk remains
- **Synchronization risk:** each arbitrageur waits for others to act first $\Rightarrow$ no one moves
- **Short-selling constraints:** borrowing shares is costly, sometimes impossible (“short squeeze”)
- **Model risk:** the apparent mispricing might reflect a missing risk factor
- **Example:** LTCM (1998) — convergence trades were “correct” but blow-up before convergence

Key Insight

- Limits to arbitrage explain why markets can be **informationally inefficient** yet **not exploitable** — reconciling EMH with behavioral anomalies

Excess Volatility Puzzle (Shiller, 1981)

Present Value Model

- ▣ Under EMH, stock price equals discounted future dividends: $P_t = \mathbb{E}_t \left[\sum_{k=1}^{\infty} \frac{D_{t+k}}{(1+r)^k} \right]$
 - ▣ Define the “ex-post rational price”: $P_t^* = \sum_{k=1}^{\infty} \frac{D_{t+k}}{(1+r)^k}$ (perfect foresight)
 - ▣ Since $P_t = \mathbb{E}_t[P_t^*]$, we must have $\text{Var}(P_t) \leq \text{Var}(P_t^*)$ (variance bound)
-
- | | |
|--|--|
| <ul style="list-style-type: none">▣ Shiller's finding:▣ S&P 500 prices far more volatile than P_t^*▣ $\sigma(P)/\sigma(P^*) \approx 5\text{--}13$ (depending on period)▣ Prices “move too much” relative to subsequent dividends | <ul style="list-style-type: none">▣ Debate:▣ Marsh & Merton (1986): dividend smoothing biases P_t^*▣ Fama (1991): time-varying risk premia explain excess volatility▣ Campbell & Shiller (1988): log-linear model confirms excess volatility▣ Remains one of the strongest challenges to EMH |
|--|--|

Sentiment Indicators

Market-Based Indicators

- ▣ **VIX** (CBOE Volatility Index, “Fear Index”): implied volatility from S&P 500 options
- ▣ High VIX $\Rightarrow$ fear $\Rightarrow$ historically predicts **higher** future returns
- ▣ **Put/Call ratio**: high ratio = bearish sentiment $\Rightarrow$ contrarian buy signal
- ▣ **Margin debt**: high margin $\Rightarrow$ euphoria $\Rightarrow$ potential correction

Baker–Wurgler Sentiment Index

- ▣ Composite of 6 proxies: closed-end fund discount, NYSE turnover, IPO volume, IPO first-day returns, equity share in new issues, dividend premium
- ▣ High sentiment $\Rightarrow$ overpricing of speculative stocks
- ▣ Low sentiment $\Rightarrow$ underpricing of speculative stocks
- ▣ Predictive for cross-section, not aggregate market

EMH Interpretation

- ▣ If sentiment predicts returns, prices do not fully reflect available information $\Rightarrow$ challenges semi-strong EMH. However, sentiment-based strategies have limited Sharpe ratios after costs.

Quiz: Behavioral Finance (Questions)

Q1

- ☐ Under prospect theory, an investor who is losing 10% on a stock is most likely to: **(A)** sell immediately **(B)** hold and hope for recovery **(C)** buy more to average down **(D)** hedge with options

Q2

- ☐ Which is NOT a limit to arbitrage? **(A)** Noise trader risk **(B)** Short-selling constraints **(C)** Diversification **(D)** Synchronization risk

Q3

- ☐ Shiller's excess volatility puzzle shows that: **(A)** dividends are too volatile **(B)** prices move more than dividends justify **(C)** volatility is constant **(D)** CAPM holds exactly

Quiz: Behavioral Finance (Answers)

A1: (B)

- ▣ Loss aversion makes investors risk-seeking in losses, so they hold losers hoping for recovery.

A2: (C)

- ▣ Diversification is a portfolio strategy, not a limit to arbitrage. Noise trader risk, short-selling constraints, and synchronization risk all prevent arbitrageurs from correcting mispricings.

A3: (B)

- ▣ Stock prices are far more volatile than the present value of subsequent dividends. Shiller showed $\sigma(P)/\sigma(P^*) \approx 5-13$, violating the variance bound implied by the EMH.

Andrew Lo's Adaptive Market Hypothesis (2004)

Core Idea

- Market efficiency is not a static property but an **evolving dynamic** driven by evolutionary forces: competition, adaptation, natural selection.
- **Key principles:**
 - Investors are neither fully rational nor fully irrational — they **adapt**
 - Market ecology: species (hedge funds, retail, High-Frequency Trading, HFT) compete for resources (alpha)
 - Profit opportunities arise and disappear as strategies become crowded
 - Risk premia vary over time with market conditions
- **Predictions:**
 - Efficiency varies across markets and time periods
 - Anomalies appear, attract capital, and then decay
 - Crises create new inefficiencies (fear, forced selling)
 - Recovery periods restore efficiency
 - EMH is a **limiting case** of AMH when market ecology is stable

EMH vs. AMH: A Comparison

Dimension	EMH (Fama)	AMH (Lo)
Investor rationality	Fully rational	Boundedly rational, adaptive
Market efficiency	Constant	Time-varying, dynamic
Risk premia	Constant / slowly varying	Fluctuate with market ecology
Anomalies	Statistical artifacts or risk	Real but transient
Crises	Rational response to news	Ecological disruptions
Investment strategy	Passive (index funds)	Adaptive, regime-dependent
Theoretical basis	Neoclassical economics	Evolutionary biology

Practical Implication

- Under AMH, the optimal strategy is **adaptive**: passive in calm periods, active during crises when inefficiencies are greatest.

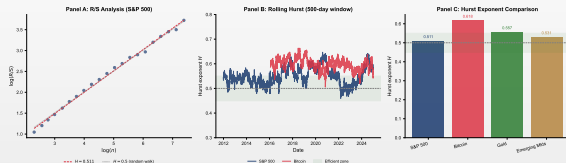
Measuring Time-Varying Efficiency

- ▣ **Rolling variance ratio**: compute $VR(q)$ over a rolling window (e.g., 500 days)
- ▣ $|VR(q) - 1|$ as a time-varying **efficiency index**
- ▣ During crises (2008, 2020), $|VR - 1|$ spikes $\Rightarrow$ temporary inefficiency
- ▣ Between crises, $|VR - 1|$ returns to low levels $\Rightarrow$ efficiency is restored

Hurst Exponent as Efficiency Measure

- ▣ $H = 0.5$: random walk (efficient)
- ▣ $H > 0.5$: persistence (trending, momentum)
- ▣ $H < 0.5$: anti-persistence (mean-reverting)
- ▣ Estimated via R/S analysis or Detrended Fluctuation Analysis (DFA)
- ▣ Typical values for daily returns:
 - ▣ S&P 500: $H \approx 0.52$
 - ▣ BTC-USD: $H \approx 0.55$ (more persistent)
 - ▣ BET: $H \approx 0.58$ (least efficient)

Hurst Exponent: R/S Analysis Plot



- Slope of $\log(R/S)$ vs. $\log(n)$ regression estimates H
- $H = 0.5$: random walk (efficient);
 $H > 0.5$: long memory; $H < 0.5$: anti-persistence
- S&P 500 line is close to the $H = 0.5$ reference — near-efficient
- BET and BTC lines have steeper slopes — persistent, less efficient
- **Caveat:** R/S analysis is biased upward for short-memory processes; DFA provides a more robust alternative

Python: Hurst Exponent via R/S Analysis

```
import numpy as np, yfinance as yf

sp = yf.download('^GSPC', start='2015-01-01', end='2025-01-01')
r = np.log(sp['AdjClose']/sp['AdjClose'].shift(1)).dropna().values

def hurst_rs(ts, min_w=20, max_w=None):
    """Estimate Hurst exponent via rescaled range."""
    N = len(ts)
    if max_w is None: max_w = N // 4
    wins = [int(w) for w in np.logspace(
        np.log10(min_w), np.log10(max_w), 20)]
    RS = []
    for w in wins:
        rv = []
        for s in range(0, N-w, w):
            seg = ts[s:s+w]; Y = np.cumsum(seg-seg.mean())
            S = seg.std(ddof=1)
            if S > 0: rv.append((Y.max()-Y.min())/S)
        RS.append((w, np.mean(rv)))
    ln, lr = np.log([x[0] for x in RS]), np.log([x[1] for x in RS])
    return np.polyfit(ln, lr, 1)[0]
print(f"Hurst (S&P500): {hurst_rs(r):.3f}") # ~0.52
```

AMH: Empirical Evidence

Time-Varying Sharpe Ratios

- ▣ Rolling 3-year Sharpe ratios for S&P 500 range from -0.5 to $+1.5$
- ▣ Momentum strategies: $SR > 1$ in 2009–2015, $SR \approx 0$ in 2018–2020
- ▣ Value strategies: profitable in 2000–2007, underperform in 2010–2020
- ▣ **No strategy is permanently profitable** — consistent with AMH

Regime-Dependent Alpha

- ▣ Rolling CAPM alpha for hedge funds varies with VIX regime
- ▣ Low VIX (< 15): $\alpha \approx 0$ (efficient)
- ▣ High VIX (> 30): $\alpha \approx 3\%–5\%$ annualized
- ▣ Crisis periods create dislocations that skilled managers exploit
- ▣ As capital flows in, alpha decays (crowding)

Key Takeaway

- ▣ The profitability of trading strategies is **cyclical**, not permanent — exactly as AMH predicts

AMH: Investment Implications

Market Regime	Optimal Strategy	Rationale
Low volatility, trending	Passive indexing	Markets are efficient; $\alpha \approx 0$
High volatility, crisis	Active/tactical	Dislocations create opportunities
Post-crisis recovery	Momentum overweight	Mean reversion from overshooting
Crowded factor regime	Factor diversification	Single factors lose edge
Structural change	Adaptive rebalancing	Old relationships break down

- ▣ **Risk parity adaptation:** equal risk contribution works in calm periods; shift to defensive in crises
- ▣ **Factor timing:** rotate between value, momentum, quality based on macro regime

- ▣ **Practical rule:** be passive by default, active only when opportunity cost of passivity is high
- ▣ **Monitoring signal:** $|VR(q) - 1|$ spike $\Rightarrow$ potential for active alpha

AMH vs Fractal Market Hypothesis (FMH)

Fractal Market Hypothesis (Peters, 1994)

- Markets are stable when investors with **different time horizons** participate
- Short-horizon traders provide liquidity to long-horizon investors
- Crises occur when all horizons **synchronize** (everyone sells)
- Multi-fractal models: MF-DFA, MF-DCCA

Comparison

	AMH	FMH
Basis	Evolution	Fractals
Key driver	Adaptation	Time horizons
Crises	Ecological	Horizon collapse
Math tool	Rolling tests	MF-DFA
Efficiency	Time-varying	Scale-dependent

Mandelbrot's Legacy

- Both AMH and FMH reject the assumption of constant market dynamics; they differ in mechanism (biological adaptation vs. mathematical self-similarity) but agree that EMH is a special case

Market Ecology

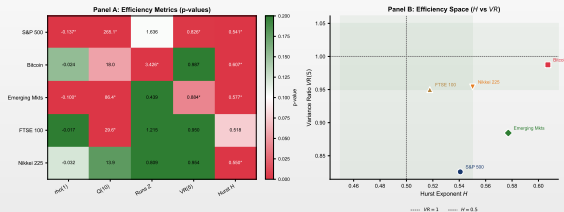
Lo's Species Analogy

- Market participants as “species” competing for alpha (“food”)
- When a niche is overcrowded, returns shrink $\Rightarrow$ weaker species exit

Species	Strategy	Impact on Efficiency
Retail investors	Buy-and-hold, sentiment-driven	Provide noise, reduce efficiency
Hedge funds	Factor-based, arbitrage	Increase efficiency (correct mispricing)
HFT firms	Market-making, latency arb	Improve short-term efficiency
Index funds	Passive replication	Price-takers; no information role
Dark pools	Off-exchange matching	Reduce transparency, mixed effect

- HFT debate:** improves liquidity and narrows spreads, but “phantom liquidity” disappears in crises
- Dark pools:** $\sim 40\%$ of US equity volume; reduce information leakage but fragment price discovery
- Passive investing growth:** if too many are passive, who corrects mispricing? (Grossman–Stiglitz paradox revisited)

Efficiency Comparison Across Markets



- Efficiency indicators (VR , H , $\hat{\rho}_1$) compared for S&P 500, FTSE 100, BET, BTC-USD
- Developed markets cluster near $VR = 1$, $H = 0.5$ — weak-form efficient
- Emerging markets (BET, SSE) show significant deviations at longer horizons
- Key drivers of efficiency:** liquidity, analyst coverage, institutional participation, regulatory quality
- BTC has converged toward developed-market efficiency since 2020 (institutional adoption, ETFs)

Case Study: Comprehensive EMH Testing on S&P 500

Setup

- Data: S&P 500 daily log-returns, Jan 2000 – Dec 2024 ($T \approx 6,300$)
- Goal: apply all weak-form tests and interpret jointly

Test	Statistic	p -value	Decision at 5%
Ljung-Box $Q(10)$	15.3	0.12	Do not reject
Ljung-Box $Q(20)$	28.1	0.11	Do not reject
Runs test	$Z = -1.42$	0.16	Do not reject
VR(2)	1.047 ($z = 1.88$)	0.06	Do not reject (marginal)
VR(5)	1.038 ($z = 1.21$)	0.23	Do not reject
VR(10)	0.993 ($z = -0.18$)	0.86	Do not reject
ADF (prices)	-1.85	0.35	Do not reject (unit root)
ADF (returns)	-52.3	< 0.001	Reject (stationary)
Hurst exponent	$H = 0.523$	—	Near 0.5 (efficient)

Case Study: Interpretation and Conclusions

Findings

- ▣ All tests fail to reject weak-form EMH at 5%
- ▣ $VR(2)$ is marginally significant — small positive autocorrelation at lag 1
- ▣ $H = 0.523 \approx 0.5 \Rightarrow$ mild persistence
- ▣ Prices: unit root; returns: stationary

Caveats

- ▣ $\hat{\rho}_1 \approx 0.05$ is significant with $T = 6,300$ but generates $< 0.1\%$ daily $\Rightarrow$ consumed by costs
- ▣ Joint hypothesis: we assume CAPM
- ▣ Subperiods around crises may reject
- ▣ $|r_t|$ is autocorrelated $\Rightarrow$ volatility is predictable, returns are not

Bottom Line

- ▣ The S&P 500 is approximately weak-form efficient. Small statistical deviations exist but are not economically exploitable after costs.

Case Study: Bitcoin Efficiency

Evolution of BTC-USD Efficiency (2015–2024)

- Bitcoin has transitioned from a clearly inefficient market to one approaching weak-form efficiency

Period	LB(10) ρ	VR(5)	$\hat{H}$	Daily vol	Efficiency
2015–2016	0.001	1.22	0.62	4.2%	Inefficient
2017 (bubble)	0.003	1.35	0.68	5.8%	Inefficient
2018–2019	0.04	1.12	0.57	3.9%	Marginal
2020–2021	0.08	1.08	0.55	3.5%	Marginal
2022–2024	0.18	1.04	0.53	2.8%	Near-efficient

- Early period:** thin liquidity, retail-dominated, strong momentum $\Rightarrow$ high H , significant autocorrelation
- Maturation:** institutional entry (CME futures 2017, ETFs 2024), increased liquidity, algorithmic trading
- Current state:** daily returns approximately efficient; intraday still shows microstructure patterns

Case Study: Bucharest Stock Exchange (BET)

BET Efficiency Results

Test	Result
LB(10) p -value	< 0.01
VR(5)	1.18 ($z = 3.2$)
$\hat{H}$ (R/S)	0.58
ADF (returns) p	< 0.001
$\hat{\rho}_1$	0.12

- Significant autocorrelation: $\hat{\rho}_1 = 0.12$
- $VR(5) = 1.18 \gg 1$: strong momentum
- $H = 0.58$: persistent, inefficient

Why Less Efficient?

- Low liquidity: avg daily turnover $\sim 10M$
- Few institutional investors
- Limited analyst coverage
- Information asymmetry between insiders and retail
- Thin order books $\Rightarrow$ large bid-ask spreads

Implications

- Momentum strategies may work *before* costs
- High transaction costs offset predictability
- EU accession improved efficiency over time

Case Study: Gold as Safe Haven

Gold Returns Properties

- ▣ Daily returns: $\mu \approx 0.03\%$, $\sigma \approx 1.0\%$
- ▣ Skewness: -0.15 (slight left skew)
- ▣ Kurtosis: 7.2 (fat tails)
- ▣ Low correlation with equities ($\rho \approx -0.05$)

Efficiency Tests

Test	Result
LB(10) p	0.35
$VR(5)$	0.97
$\hat{H}$	0.49

- ▣ $VR(5) = 0.97 < 1$: slight mean reversion
- ▣ $H = 0.49 < 0.5$: anti-persistent
- ▣ Gold is highly liquid (24/7 OTC market) $\Rightarrow$ near-efficient
- ▣ **Safe haven property**: negative correlation with equities during crises (2008, 2020)
- ▣ **Mean reversion**: gold tends to revert to production cost / inflation-adjusted mean
- ▣ Not a pure random walk, but efficient in practice (deviations too small to exploit)

Cross-Asset Efficiency Ranking

Rank	Asset	$ VR(5) - 1 $	$ \hat{H} - 0.5 $	$ \hat{\rho}_1 $	Efficiency Score
1	EUR/USD (FX)	0.01	0.01	0.01	0.99
2	S&P 500	0.04	0.02	0.05	0.96
3	Gold (XAU)	0.03	0.01	0.03	0.97
4	US 10Y Treasury	0.05	0.03	0.04	0.95
5	BTC-USD	0.04	0.03	0.05	0.95
6	BSE Sensex	0.08	0.04	0.07	0.90
7	BET (Romania)	0.18	0.08	0.12	0.80
8	SSE (China)	0.15	0.07	0.10	0.83

- **Efficiency score:** $1 - \frac{1}{3}(|VR - 1| + 2|\hat{H} - 0.5| + |\hat{\rho}_1|)$, normalized to $[0, 1]$
- FX markets are most efficient: highest liquidity, 24-hour trading, most participants
- Emerging equity markets are least efficient: lower liquidity, fewer analysts, more retail
- BTC has improved dramatically; now comparable to developed market equities

Event Study: AAPL Earnings Announcement



- **Event:** Apple Q4 2024 earnings beat (+8% EPS surprise)
- $\overline{AR}_0 = +3.2\%$ on announcement day (statistically significant, $t = 4.8$)
- $\overline{CAR}(-1, +1) = +4.1\%$: most adjustment within 1 day
- $\overline{CAR}(+2, +20) = +0.8\%$: small post-event drift (mild PEAD)
- **Interpretation:** mostly consistent with semi-strong EMH; the bulk of adjustment is immediate, with only a small residual drift

Time-Varying Efficiency: Evidence



- ▣ Rolling $|VR(5) - 1|$ (500-day window) for S&P 500 spikes during crises:
- ▣ **GFC (2008–09):** $|VR - 1| \approx 0.12$ — autocorrelation from panic selling, forced liquidation
- ▣ **COVID (2020):** $|VR - 1| \approx 0.08$ — brief inefficiency, rapid recovery
- ▣ **Calm periods:** $|VR - 1| < 0.03$ — near-perfect random walk
- ▣ **Rolling H :** spikes to ~ 0.60 during crises (persistence), returns to ~ 0.52 in calm periods
- ▣ Crisis-driven inefficiency creates **temporary alpha opportunities** for adaptive investors (AMH)

Efficiency and Market Microstructure

Microstructure Effects on Tests

- ▣ **Bid-ask bounce:** trade prices alternate between bid and ask $\Rightarrow$ negative autocorrelation ($\hat{\rho}_1 < 0$) even if true returns are i.i.d.
- ▣ Roll (1984): $\hat{\rho}_1 \approx -s^2/4$ where s is the spread
- ▣ **Non-synchronous trading:** illiquid stocks update prices with delay $\Rightarrow$ spurious cross-autocorrelation

Epps Effect

- ▣ Correlations between assets **decrease** as sampling frequency increases
- ▣ At 1-min: $\rho \approx 0.15$; at daily: $\rho \approx 0.50$
- ▣ Caused by non-synchronous arrivals of information
- ▣ Implication: high-frequency tests of EMH are contaminated by microstructure noise

Practical Rule

- ▣ Use daily or lower frequency for EMH tests to avoid microstructure artifacts

Summary: The State of EMH

What the evidence supports

- ▣ Most actively managed funds **underperform** passive benchmarks
- ▣ Many anomalies **decay** after publication
- ▣ Short-term price prediction is extremely difficult
- ▣ Transaction costs eliminate most trading profits

What challenges EMH

- ▣ Momentum and value effects persist
- ▣ PEAD is robust across markets
- ▣ Insider trading is profitable
- ▣ Bubbles and crashes occur periodically
- ▣ Market efficiency varies over time

Practical Takeaway

- ▣ Markets are “efficiently inefficient” (Pedersen, 2015): efficient enough that most cannot beat them, but with enough friction to reward sophisticated, informed investors.

AI Exercise: Testing Market Efficiency

Prompt to test in ChatGPT / Claude / Copilot

“Download daily data for S&P 500 from 2015 to 2025 using Python (yfinance). Compute log-returns. Test the weak-form EMH using: (1) Ljung–Box test on the first 20 lags, (2) Lo–MacKinlay variance ratio test for $q = 2, 5, 10, 20$, (3) Hurst exponent via R/S analysis, (4) ADF unit root test on both prices and returns. Interpret each result. Is the S&P 500 weak-form efficient?”

Exercise:

1. Does the AI compute log-returns correctly? Check: $r_t = \ln(P_t/P_{t-1})$, not simple returns
2. Is the Ljung–Box test applied to **returns** (not prices)? Does it use the correct $\chi^2(m)$ distribution?
3. Does the variance ratio correctly use $VR(q) = \text{Var}(r_t(q))/(q \cdot \text{Var}(r_t))$? Check the formula
4. Is the Hurst exponent computed via R/S or DFA? Is $H \approx 0.5$ interpreted as efficient?
5. Does the AI distinguish between **statistical** and **economic** significance?
6. Ask: “What if I test BET (Romania) instead?” Does the AI recognize emerging market differences?

❑ **Warning:** AI often applies the ADF test to *returns* and concludes “no unit root = efficient.” This is wrong — the ADF on returns tests stationarity, not efficiency. The relevant test is on *prices*.

Question 1

Question

- ▣ A researcher computes the Ljung–Box statistic $Q(10) = 18.5$ for daily S&P 500 returns. The critical value $\chi^2_{10,0.05} = 18.31$. What is the conclusion?

Answer choices

- ▣ **(A)** Reject weak-form EMH: returns are autocorrelated
- ▣ **(B)** Do not reject EMH: the test is inconclusive
- ▣ **(C)** Reject strong-form EMH
- ▣ **(D)** Cannot decide without the variance ratio

Question 1: Answer

Answer: (A)

- ▣ Since $Q(10) = 18.5 > 18.31 = \chi_{10,0.05}^2$, we reject H_0 at 5%.
- ▣ H_0 : $\rho_1 = \dots = \rho_{10} = 0$ (returns are uncorrelated).
- ▣ Rejection means returns have statistically significant autocorrelation $\Rightarrow$ challenges weak-form EMH.
- ▣ **Caveat:** statistical significance $\neq$ economic significance. The autocorrelations may be too small to exploit profitably after transaction costs.
- ▣ (B) is wrong: $Q > \chi_{\text{crit}}^2 \Rightarrow$ we *do* reject. (C) is wrong: Ljung–Box tests weak-form, not strong-form.

Question 2

Question

- ☐ A market has Hurst exponent $H = 0.65$. This implies:

Answer choices

- ☐ (A) Returns are anti-persistent (mean-reverting)
- ☐ (B) Returns are consistent with a random walk
- ☐ (C) Returns are persistent (trending behavior)
- ☐ (D) The market is strong-form efficient

Question 2: Answer

Answer: (C)

- ▣ $H > 0.5$ indicates persistent behavior: positive returns tend to follow positive returns (and vice versa).
- ▣ $H = 0.5$: random walk (no memory). $H < 0.5$: anti-persistent (mean-reverting). $H > 0.5$: persistent (trending).
- ▣ $H = 0.65$ is well above 0.5, suggesting significant long-range dependence.
- ▣ This challenges weak-form EMH: past returns contain information about future returns.
- ▣ (A) would require $H < 0.5$. (B) requires $H = 0.5$. (D) is unrelated to the Hurst exponent.

Question 3: Reasoning — Event Study Interpretation

Question

- After a positive earnings surprise, a stock's CAR drifts upward by 2% over the following 60 days. This is evidence against:

Answer choices

- (A) Weak-form efficiency only
- (B) Semi-strong efficiency (the information was public)
- (C) Strong-form efficiency only
- (D) No form of efficiency — anomalies are always explained by risk

Question 3: Answer

Answer: (B)

- ▣ Earnings announcements are **public information**. If the market is semi-strong efficient, prices should adjust immediately and completely.
- ▣ A 60-day drift after the announcement means the market **underreacts** to public information $\Rightarrow$ violation of semi-strong EMH.
- ▣ This is the Post-Earnings Announcement Drift (PEAD), one of the most robust anomalies.
- ▣ (A) is incomplete: PEAD uses public info, not just past prices. (C) is wrong: insider info is not involved.
- ▣ (D) is Fama's counter-argument, but the drift pattern is systematic and one-directional.

Exercise 1: Ljung–Box Test and Weak-Form EMH

Problem

▣ A researcher computes the following sample autocorrelations from $T = 500$ daily stock returns:

Lag k	1	2	3	4	5
$\hat{\rho}_k$	0.08	-0.03	0.05	0.02	-0.01

1. Compute the Ljung–Box statistic: $Q(5) = T(T+2) \sum_{k=1}^5 \frac{\hat{\rho}_k^2}{T-k}$
2. Compare $Q(5)$ with $\chi_5^2(0.05) = 11.07$. Is weak-form EMH rejected?
3. Is $\hat{\rho}_1 = 0.08$ individually significant? (Use $z_{\alpha/2}/\sqrt{T}$ with $z_{0.025} = 1.96$)
4. Discuss: if $\hat{\rho}_1$ is significant but $Q(5)$ is not, what does this mean economically?

Exercise 2: Variance Ratio Test

Problem

□ For a stock index, you estimate: $\hat{\sigma}^2(r_t) = 0.0001$ (daily variance) and $\hat{\sigma}^2(r_t(5)) = 0.00058$ (5-day return variance), with $T = 1,000$ observations.

1. Compute $VR(5) = \hat{\sigma}^2(r_t(5)) / (5 \cdot \hat{\sigma}^2(r_t))$
2. Is $VR(5) > 1$ or < 1 ? Interpret in terms of autocorrelation (momentum vs. mean reversion)
3. Compute the asymptotic variance under homoscedasticity: $\phi(5) = \frac{2(2 \cdot 5 - 1)(5 - 1)}{3 \cdot 5 \cdot T}$
4. Compute the z-statistic: $z = (VR(5) - 1) / \sqrt{\phi(5)}$. Is the random walk rejected at 5%?

Hint

$z_{0.025} = 1.96$. Reject H_0 (random walk) if $|z| > 1.96$.

Exercise 3: Event Study

Problem

- An event study examines $N = 30$ firms around earnings announcements. Market model parameters: $\hat{\alpha} = 0.001$, $\hat{\beta} = 1.2$. On the event day:

	Firm return R_i	Market return R_m
Average across 30 firms	2.5%	0.3%

1. Compute the average abnormal return: $\overline{AR} = \bar{R}_i - (\hat{\alpha} + \hat{\beta}\bar{R}_m)$
2. If the standard deviation of AR_i across firms is $\hat{\sigma}_{AR} = 1.8\%$, compute the t -statistic:
 $t = \overline{AR}/(\hat{\sigma}_{AR}/\sqrt{N})$
3. Is the abnormal return significant at 5% ($t_{0.025,29} \approx 2.045$)?
4. If the $\overline{CAR}(-10, +10) = 3.2\%$ with $\hat{\sigma}_{CAR} = 4.5\%$, is the cumulative effect significant?

Exercise 4: ADF Test Interpretation

Problem

- ▣ A researcher applies the ADF test (with constant, no trend) to daily log-prices of a stock index with $T = 2,000$ observations. The results are:

ADF statistic (prices)	-2.15
ADF statistic (returns)	-38.7
Critical value (5%, no trend)	-2.86
Optimal lag (BIC)	$p = 3$

1. Is there a unit root in log-prices? Compare the test statistic with the critical value.
2. Are log-returns stationary? Interpret the ADF on returns.
3. What does this jointly imply about weak-form EMH?
4. If the KPSS test on prices gives statistic = 0.52 with 5% critical value = 0.146, does this confirm or contradict the ADF result?

Exercise 5: Filter Rule Profitability

Problem

- ▣ A stock has the following daily prices over 10 days: 100, 101.5, 100.8, 99.5, 100.2, 101.8, 103.0, 102.1, 100.5, 101.0. Transaction cost is 0.2% per trade (round-trip 0.4%).
1. Apply a 1% filter rule: buy when price rises 1% from a trough; sell when it falls 1% from a peak. List all trades.
 2. Compute the total return from the filter rule (before and after costs).
 3. Compute the buy-and-hold return over the same period.
 4. Does the filter rule outperform buy-and-hold? What does this imply for weak-form EMH?

Hint

Track running min (trough) and running max (peak). Buy signal: $P_t/\text{trough} \geq 1.01$. Sell signal: $P_t/\text{peak} \leq 0.99$.

Exercise 6: Hurst Exponent Interpretation

Problem

- ▣ Four markets have the following estimated Hurst exponents (via R/S analysis on daily returns):

Market	$\hat{H}$
Market A	0.48
Market B	0.52
Market C	0.61
Market D	0.55

1. Rank these markets from most efficient to least efficient.
2. Which market(s) exhibit mean reversion? Which exhibit persistence?
3. For Market C with $H = 0.61$, what type of trading strategy might be profitable (momentum or mean reversion)?
4. If Market A is an FX pair and Market C is an emerging equity index, is this pattern expected? Why?

Exercise 7: Event Study Interpretation

Problem

- ▣ An event study of $N = 50$ dividend increase announcements yields:

Window	$\overline{CAR}$	t -stat
$(-1, 0)$	+1.8%	4.2
$(+1, +5)$	+0.9%	1.6
$(+1, +30)$	+2.5%	2.8
$(+1, +60)$	+3.1%	2.1

1. Is the immediate reaction $\overline{CAR}(-1, 0)$ significant at 5%? ($t_{0.025, 49} \approx 2.01$)
2. Is the 5-day post-event drift significant? What about the 30-day drift?
3. Does semi-strong EMH hold? Justify using the CAR pattern.
4. Propose a trading strategy to exploit this pattern. What are the risks?

AI Exercise 2: Behavioral Finance Analysis

Prompt to test in ChatGPT / Claude / Copilot

“Using Python, simulate a portfolio of 20 stocks over 1 year with daily returns drawn from $\mathcal{N}(0.0003, 0.02^2)$. Implement an investor who follows the disposition effect: sells any stock that has gained 5% from purchase price, but holds any stock that has lost more than 5%. Compare this disposition-effect portfolio’s terminal value with a buy-and-hold portfolio. Run 1000 simulations and report the average underperformance.”

Verification checklist:

1. Does the AI correctly track each stock’s purchase price (reference point)?
2. Does it implement the asymmetric selling rule correctly (sell winners at +5%, hold losers below −5%)?
3. After selling a winner, does the AI reinvest the proceeds? In what?
4. Does the simulation account for transaction costs on each sale?
5. Is the comparison fair? (Same random seed, same initial portfolio)
6. Ask: “Now add loss aversion with $\lambda = 2.25$ to the investor’s utility. How does this change the holding pattern?”

☐ **Warning:** AI may forget to track the reference point after reinvestment, or may apply symmetric

Quiz: Comprehensive EMH (Questions)

Questions

1. A market where prices reflect all public and private information is: **(A)** weak-form efficient **(B)** semi-strong efficient **(C)** strong-form efficient **(D)** informationally complete
2. The joint hypothesis problem means: **(A)** two assets are always correlated **(B)** efficiency cannot be tested without an equilibrium model **(C)** all anomalies are real **(D)** markets are always efficient
3. If $VR(5) = 0.85$ with $z = -2.5$, returns exhibit: **(A)** positive autocorrelation **(B)** negative autocorrelation (mean reversion) **(C)** random walk **(D)** unit root
4. Under the AMH, market efficiency is: **(A)** constant across time **(B)** always increasing **(C)** time-varying and regime-dependent **(D)** irrelevant
5. The KPSS test differs from ADF because its null hypothesis is: **(A)** unit root **(B)** stationarity **(C)** normality **(D)** no autocorrelation
6. Post-Earnings Announcement Drift (PEAD) challenges: **(A)** weak-form EMH **(B)** semi-strong EMH **(C)** strong-form EMH **(D)** CAPM only

Quiz: Comprehensive EMH (Answers)

Answers

- ▣ **Q1: (C)** — Strong-form efficiency incorporates all information including private/insider information.
- ▣ **Q2: (B)** — Any test of efficiency requires an asset pricing model; rejection may reflect model failure, not inefficiency.
- ▣ **Q3: (B)** — $VR < 1$ with significant z-statistic indicates negative autocorrelation (mean reversion). Returns that go up tend to go down next.
- ▣ **Q4: (C)** — The AMH posits that efficiency varies with market conditions, investor ecology, and evolutionary pressures.
- ▣ **Q5: (B)** — KPSS reverses the null: H_0 is stationarity, while ADF and PP test H_0 : unit root.
- ▣ **Q6: (B)** — PEAD involves slow adjustment to public information (earnings), which violates semi-strong efficiency.

Python Lab — Interactive Exercises

Lab 1: Weak-Form EMH Testing (Y3)

1. Download S&P 500 and BET data (2015–2025); compute log-returns
2. Plot ACF for returns and squared returns
3. Ljung–Box test for lags 10 and 20; interpret p -values
4. Compute variance ratio for $q = 2, 5, 10, 20$

Lab 2: Advanced EMH Analysis (MSc)

1. Compute Hurst exponent via R/S analysis for 5 markets
2. Rolling variance ratio (500-day window) on S&P 500
3. Event study simulation: generate abnormal returns, compute CAR
4. Compare efficiency across developed and emerging markets

Python: Technical Trading Backtest

```
import yfinance as yf, numpy as np, pandas as pd

sp = yf.download('^GSPC', start='2015-01-01', end='2025-01-01')
p = sp['AdjClose']; r = np.log(p/p.shift(1)).dropna()

# Moving average crossover: MA(50) vs MA(200)
ma50 = p.rolling(50).mean(); ma200 = p.rolling(200).mean()
signal = (ma50 > ma200).astype(int).shift(1) # 1=long, 0=cash
signal = signal.reindex(r.index).fillna(0)

strat_r = signal * r # strategy returns
bh_r = r # buy-and-hold returns

cum_strat = np.exp(strat_r.cumsum()).iloc[-1] - 1
cum_bh = np.exp(bh_r.cumsum()).iloc[-1] - 1
sr_strat = strat_r.mean()/strat_r.std()*np.sqrt(252)
sr_bh = bh_r.mean()/bh_r.std()*np.sqrt(252)

print(f"MA crossover: {cum_strat:.1%}, SR={sr_strat:.2f}")
print(f"Buy & hold: {cum_bh:.1%}, SR={sr_bh:.2f}")
# Typically: MA crossover underperforms buy-and-hold
```

Python: Sentiment Analysis (VIX vs Future Returns)

```
import yfinance as yf, numpy as np
from scipy import stats

sp = yf.download('^GSPC', start='2010-01-01', end='2025-01-01')
vix = yf.download('^VIX', start='2010-01-01', end='2025-01-01')

# Align dates
common = sp.index.intersection(vix.index)
sp_r = np.log(sp['AdjClose']/sp['AdjClose'].shift(1)).loc[common]

# Forward 20-day return vs current VIX
fwd_20d = sp_r.rolling(20).sum().shift(-20).dropna()
vix_lvl = vix['AdjClose'].loc[fwd_20d.index]

slope, intercept, r_val, p_val, _ = stats.linregress(
    vix_lvl.values.flatten(), fwd_20d.values.flatten())
print(f"Slope: {slope:.4f}, R^2: {r_val**2:.3f}, p: {p_val:.4f}")
# High VIX predicts higher future returns (contrarian signal)
# R^2 is small (~2-5%) but statistically significant
```

Python: Cross-Asset Rolling Efficiency

```
import yfinance as yf, numpy as np, pandas as pd
import matplotlib.pyplot as plt

tickers = {'^GSPC': 'S&P500', 'GC=F': 'Gold', 'BTC-USD': 'BTC'}
data = {}
for tk, name in tickers.items():
    d = yf.download(tk, start='2018-01-01', end='2025-01-01')
    data[name] = np.log(d['AdjClose']/d['AdjClose'].shift(1))

def rolling_vr(r, q=5, window=500):
    out = pd.Series(index=r.index, dtype=float)
    for i in range(window, len(r)):
        seg = r.iloc[i-window:i].dropna().values
        var1 = np.var(seg, ddof=1)
        rq = np.array([seg[j:j+q].sum() for j in range(len(seg)-q)])
        out.iloc[i] = np.var(rq, ddof=1)/(q*var1) if var1>0 else np.nan
    return out

fig, ax = plt.subplots(figsize=(10, 4))
for name, r in data.items():
    vr = rolling_vr(r.dropna())
    ax.plot(vr.index, abs(vr-1), label=name, alpha=0.7)
ax.legend(); ax.set_ylabel('|VR(5)-1|')
plt.savefig('ch3_rolling_efficiency.pdf', bbox_inches='tight')
```

Exercise 1: Solution

Solution

1. $Q(5) = 500 \times 502 \times \left(\frac{0.08^2}{499} + \frac{0.03^2}{498} + \frac{0.05^2}{497} + \frac{0.02^2}{496} + \frac{0.01^2}{495} \right) =$
 $251,000 \times (0.01282 + 0.00181 + 0.00503 + 0.00081 + 0.00002)/1000$
 $= 251,000 \times 0.00002049 \approx 251,000 \times 2.049 \times 10^{-5} \approx 5.14$
2. $Q(5) = 5.14 < 11.07 = \chi_5^2(0.05)$: **do not reject** weak-form EMH
3. Individual test: $|\hat{\rho}_1| = 0.08 > 1.96/\sqrt{500} = 0.0877$? No, $0.08 < 0.0877 \Rightarrow$ **not significant** individually either
4. Even if $\hat{\rho}_1$ were significant, a daily autocorrelation of 0.08 generates expected profit $\approx 0.08 \times \sigma \approx 0.08\%$ per day — likely consumed by bid-ask spread (~ 0.05 – 0.10%)

Exercise 2: Solution

Solution

1. $VR(5) = 0.00058 / (5 \times 0.0001) = 0.00058 / 0.0005 = 1.16$
2. $VR(5) = 1.16 > 1 \Rightarrow$ positive autocorrelation $\Rightarrow$ **momentum** (short-term trending)
3. $\phi(5) = \frac{2 \times 9 \times 4}{3 \times 5 \times 1000} = \frac{72}{15000} = 0.0048$; $\sqrt{\phi(5)} = 0.0693$
4. $z = (1.16 - 1) / 0.0693 = 0.16 / 0.0693 = 2.31$. Since $|z| = 2.31 > 1.96$: **reject** the random walk at 5%. The positive autocorrelation is statistically significant.

Interpretation

- $VR(5) = 1.16$ implies average autocorrelation $\bar{\rho} \approx (VR - 1) \times q / (2(q - 1)) = 0.16 \times 5 / 8 = 0.10$. This is typical for emerging markets and suggests potential inefficiency.

Exercise 3: Solution

Solution

1. $\overline{AR} = 0.025 - (0.001 + 1.2 \times 0.003) = 0.025 - 0.0046 = 0.0204 = 2.04\%$
2. $t = 0.0204 / (0.018 / \sqrt{30}) = 0.0204 / 0.003286 = 6.21$
3. $|t| = 6.21 > 2.045 \Rightarrow$ **significant** at 5%. The abnormal return on the event day is highly significant.
4. $t_{CAR} = 0.032 / (0.045 / \sqrt{30}) = 0.032 / 0.00822 = 3.89 > 2.045 \Rightarrow$ **significant**. The cumulative post-event drift is also significant, suggesting the market underreacts to earnings news (PEAD).

Discussion

- The fact that $\overline{CAR}(-10, +10) = 3.2\% > \overline{AR}_0 = 2.04\%$ means the market continued to drift **after** the announcement, violating semi-strong EMH.

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Key Takeaways — What to Remember from This Chapter

5 fundamental lessons

1. EMH is a **joint hypothesis**: efficiency + equilibrium model. Rejection may reflect either.
2. Weak-form tests (autocorrelation, VR, Hurst) generally support efficiency in **developed** markets but reject it in **emerging** markets
3. Event studies reveal robust anomalies (PEAD) that challenge semi-strong efficiency
4. Market efficiency is **not constant**: it varies over time and across markets (AMH)
5. **Statistical significance** $\neq$ **economic significance**: small autocorrelations may be real but unexploitable after costs

Preparation for upcoming chapters

- ▣ **Ch. 4:** Extreme Value Theory and Copulas — tail dependence and risk
- ▣ **Ch. 5:** Volatility estimation — GARCH, realized volatility

Thank You! Questions? Course materials available at:

<https://danpele.github.io/SFM/>  Quantlet  Quantinar